

Amulet Motion Controller

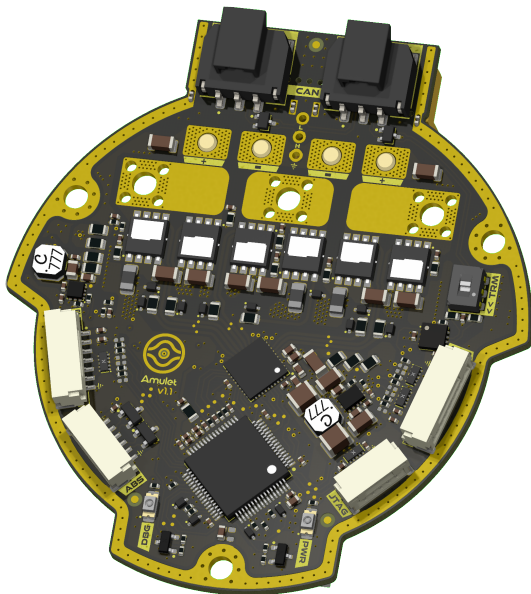
Variant: Released

2024-11-15

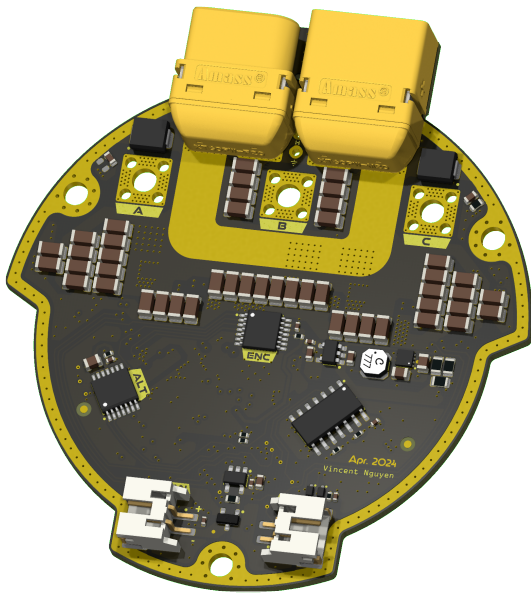
Rev 1.1

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

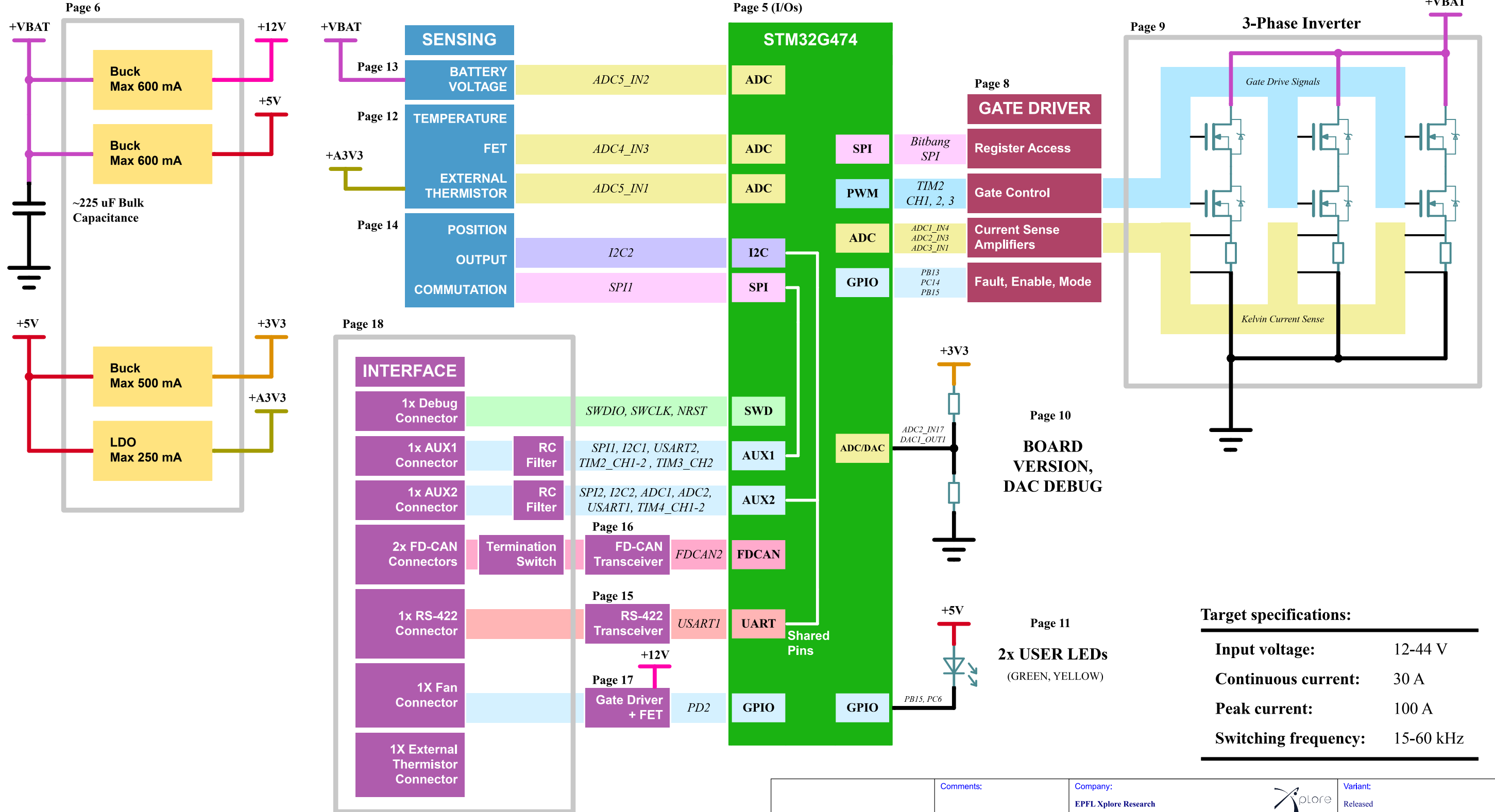
NOTES

- Schematic based off Josh Pieper's moteus controllers.
- Not fitted components are marked as **X**
- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Released 13-APR-2024

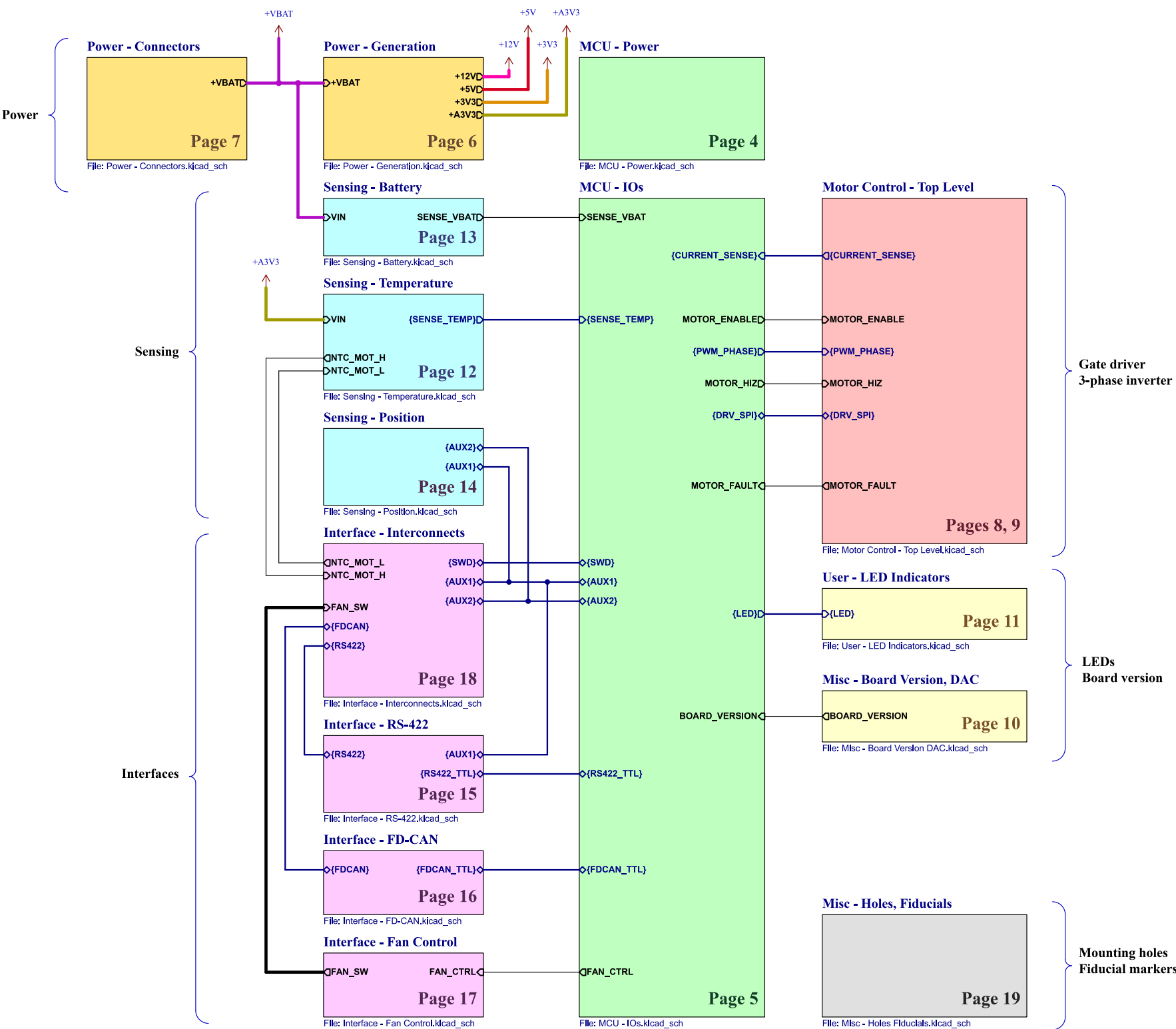
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	Sheet Title: Cover Page	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Reviewer:	Revision: 1.1
			Size: A3	Sheet: 1 of 21

[2] Block Diagram



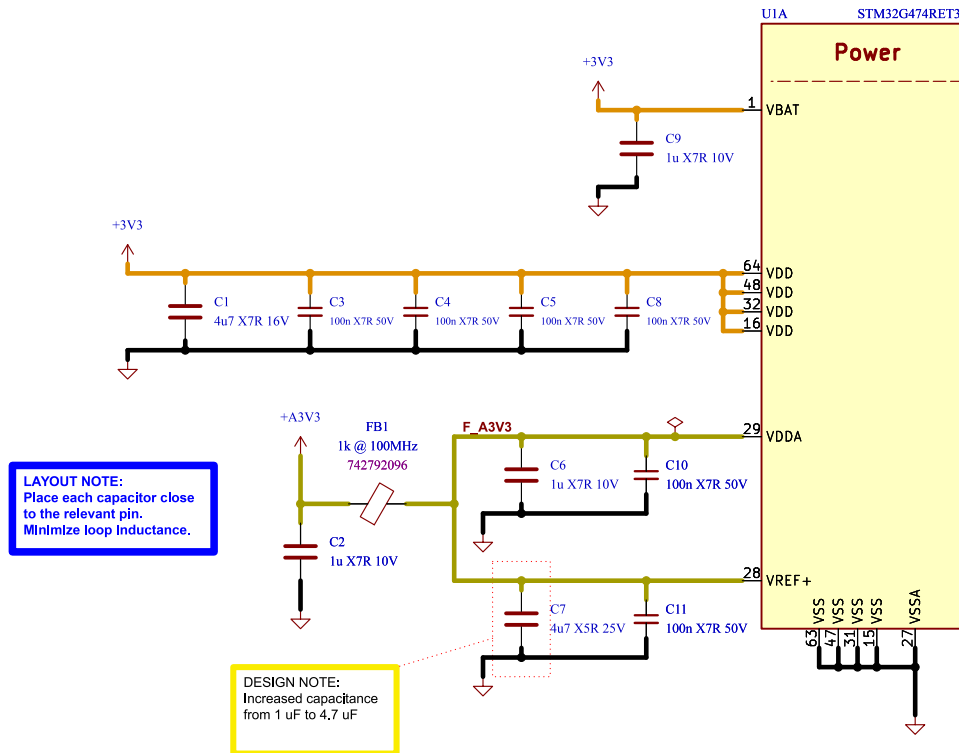
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	Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1	
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21	

[3] Project Architecture



Comments:	Company: EPFL Xplore Research				Variant: Released	
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1		
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 21		

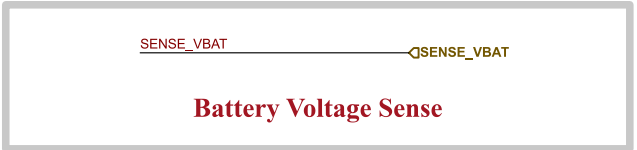
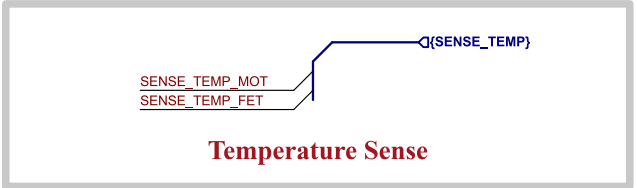
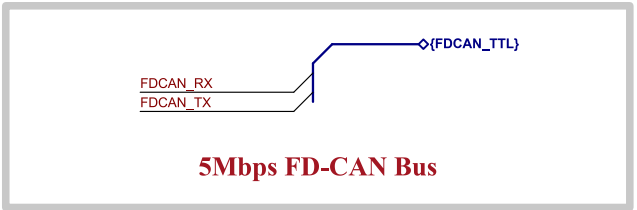
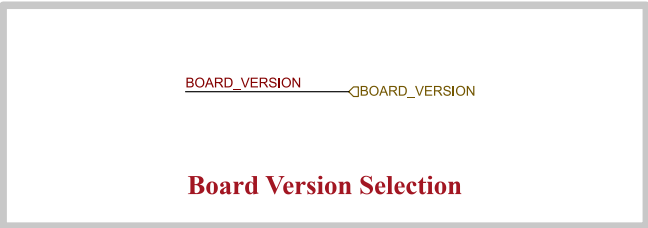
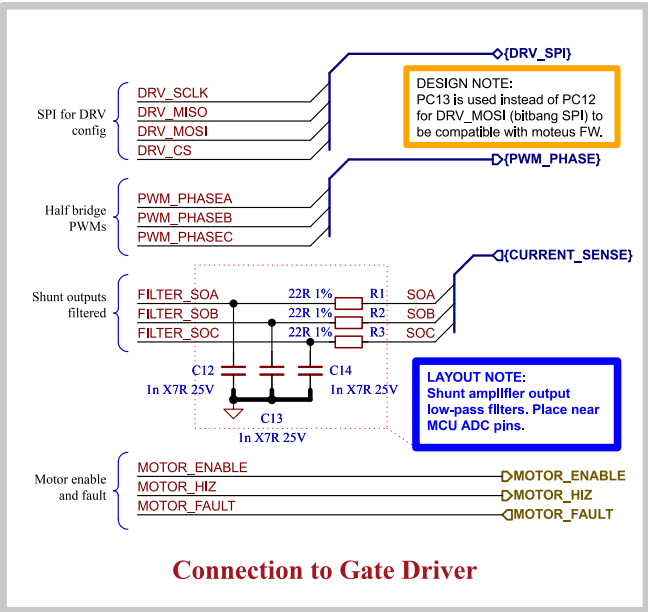
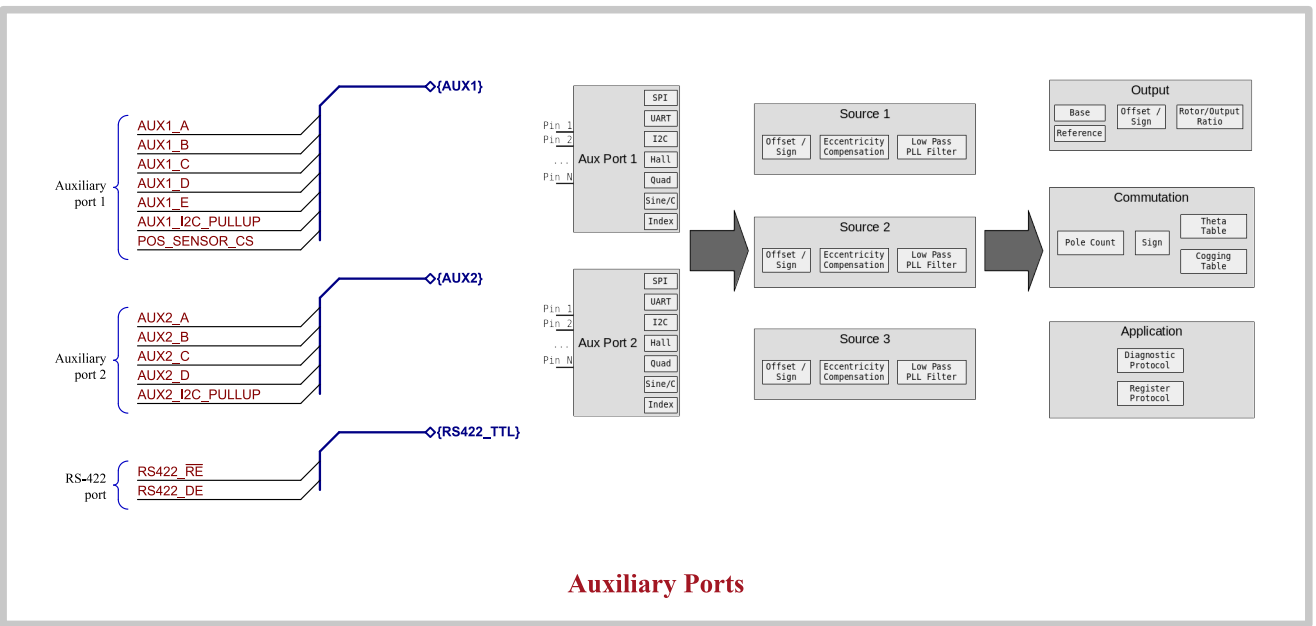
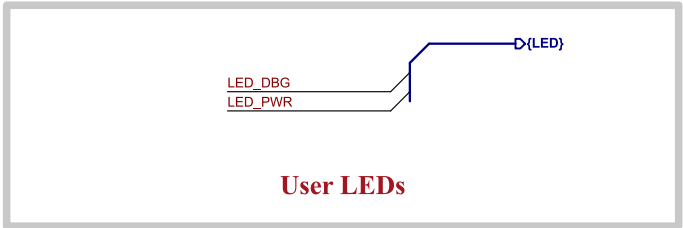
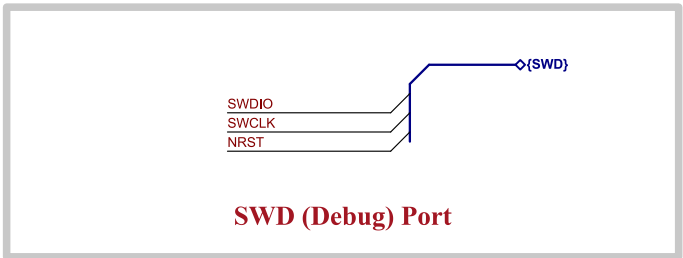
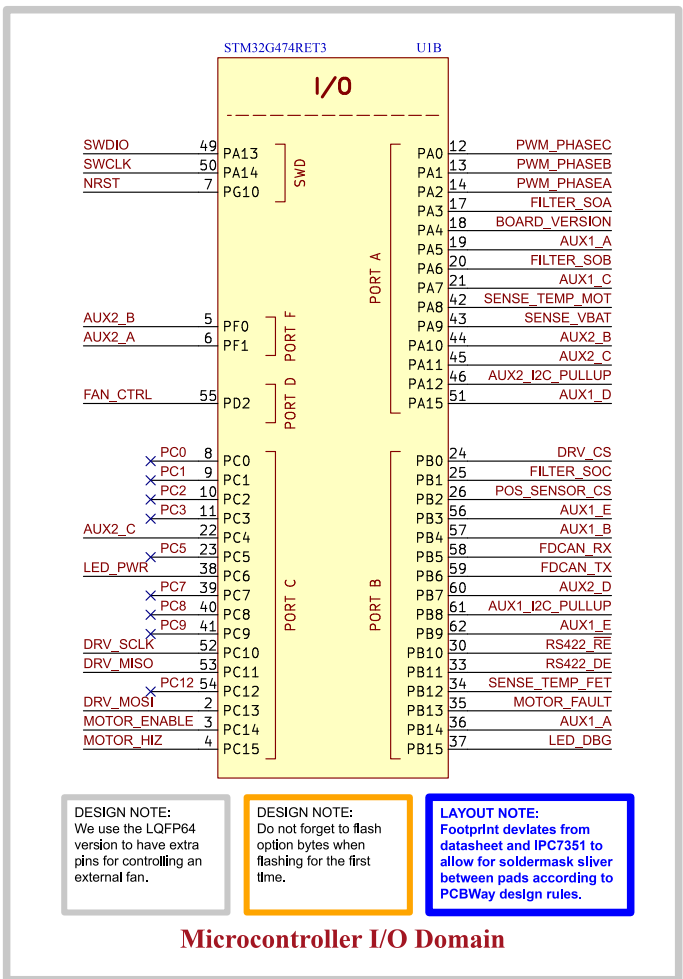
[4] MCU - Power



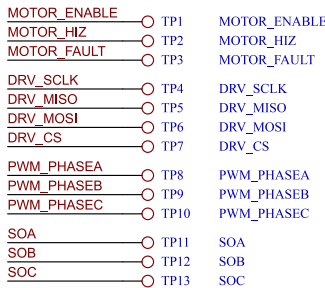
Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: EPFL Xplore Research 	Variant: Released		
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé		
	Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A4	Sheet: 4 of 21	

[5] MCU - I/Os



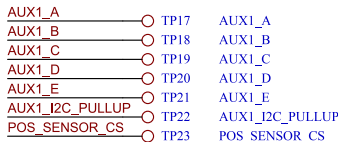
Gate Driver



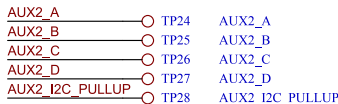
Debug



Auxiliary pins 1



Auxiliary pins 2



RS-422



LEDs



FD-CAN



Fan



Sense



Test Points

Comments:

References:

Flexible I/O worked examples
Flexible I/O source configuration

Company:

EPFL Xplore Research



Variant:

Released

Board Name:

Amulet Motion Controller

Project Name:

Chienpanzé

Sheet Title:

MCU - I/Os

File Name:

MCU - IOs.kicad_sch

Designer:

Vincent Nguyen

Date:

2023-12-20

Revision:

1.1

Sheet Path:

/Project Architecture/MCU - IOs/

Reviewer:

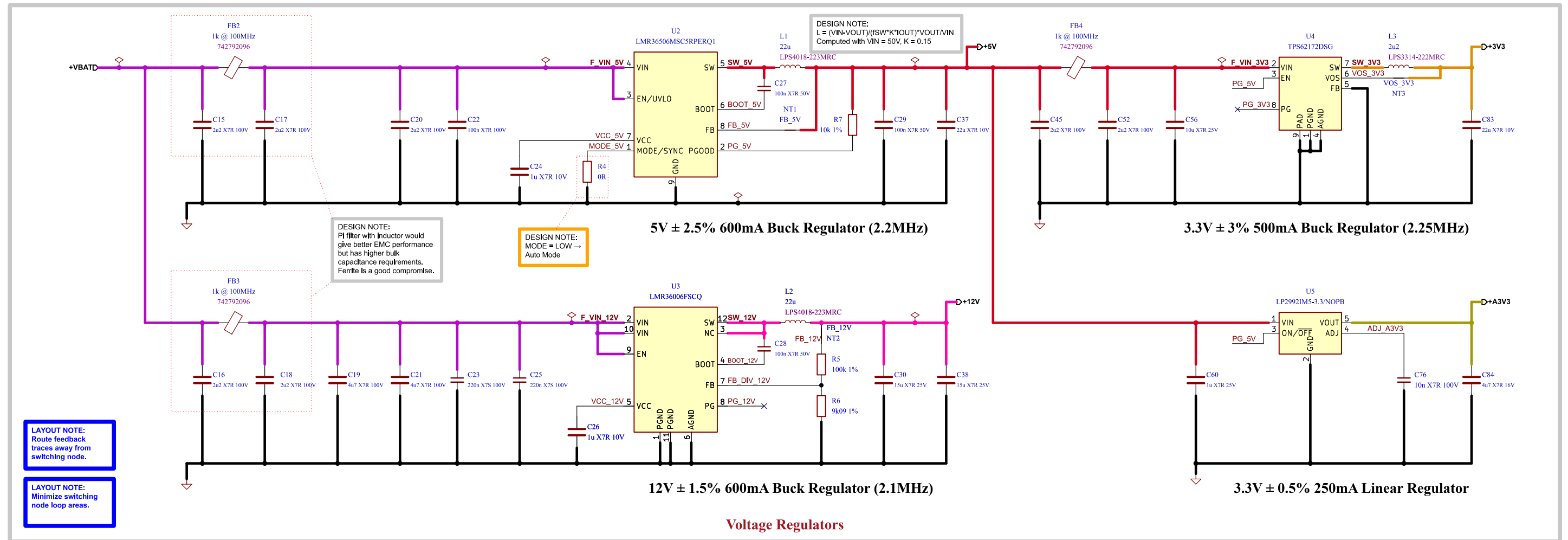
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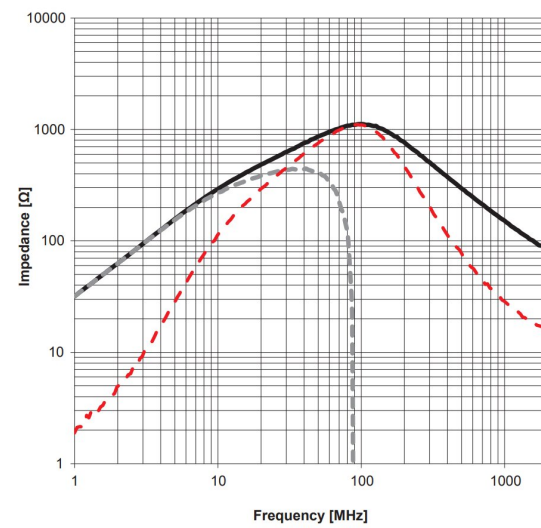
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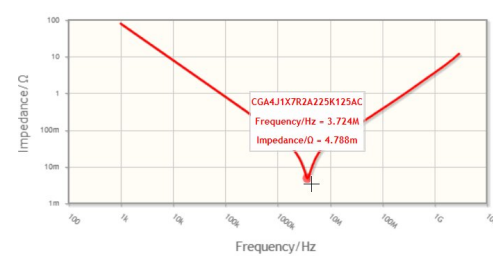
[6] Power - Generation



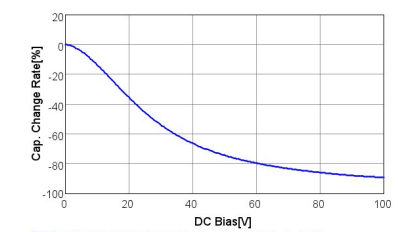
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

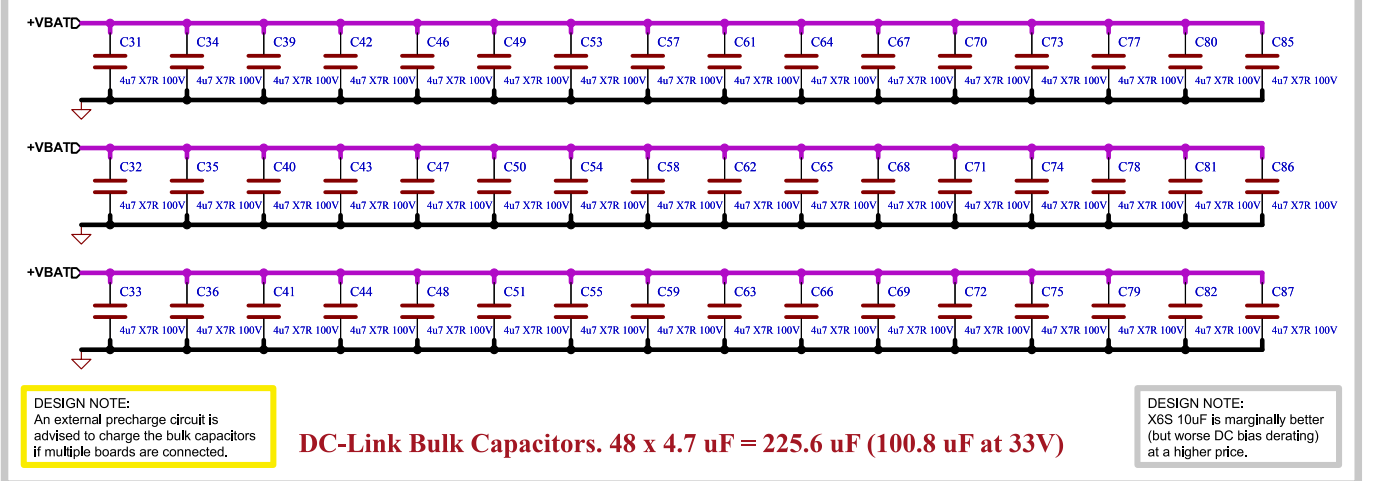
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

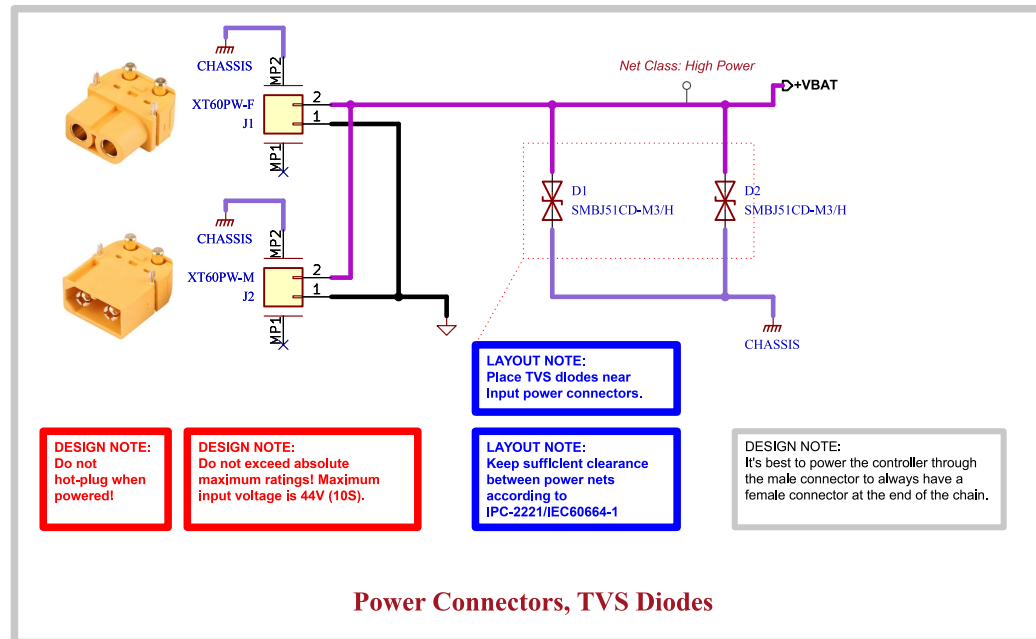
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



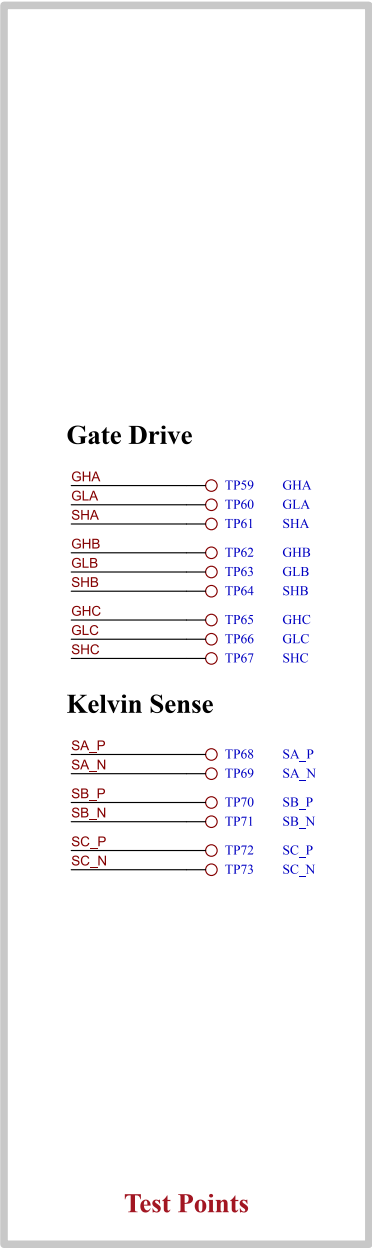
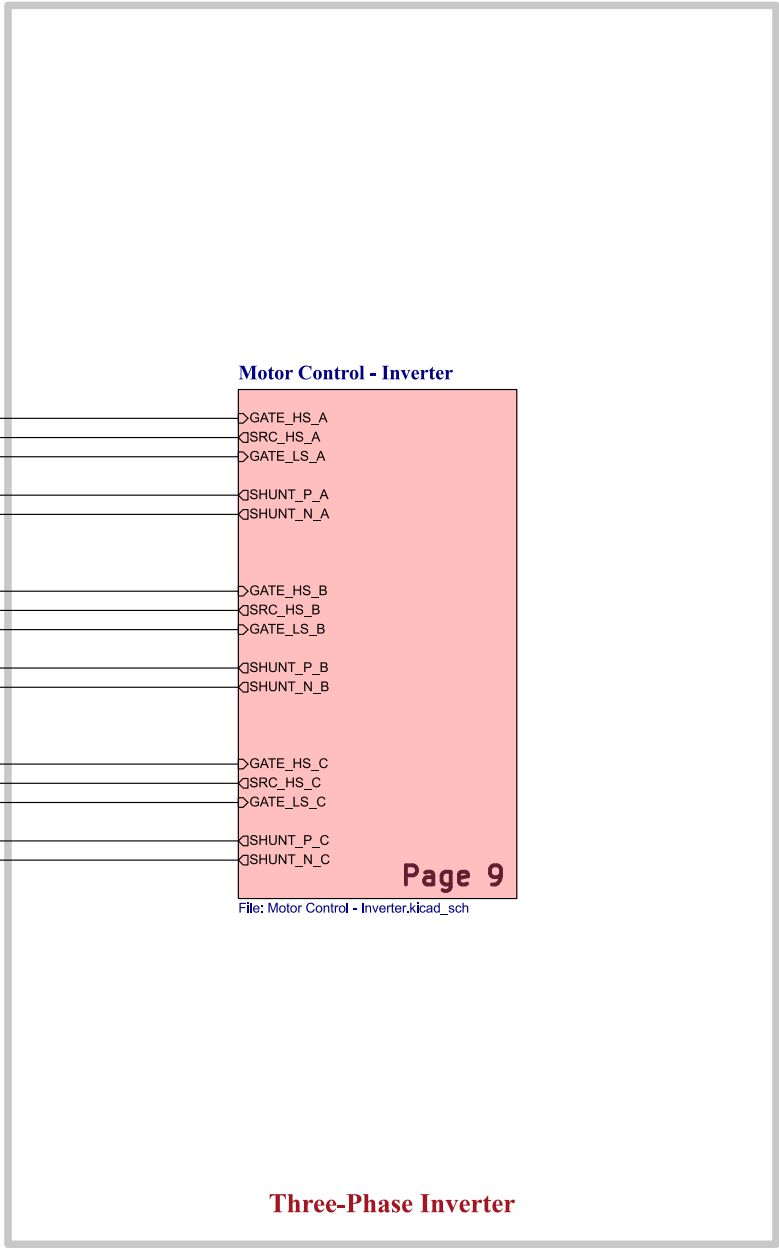
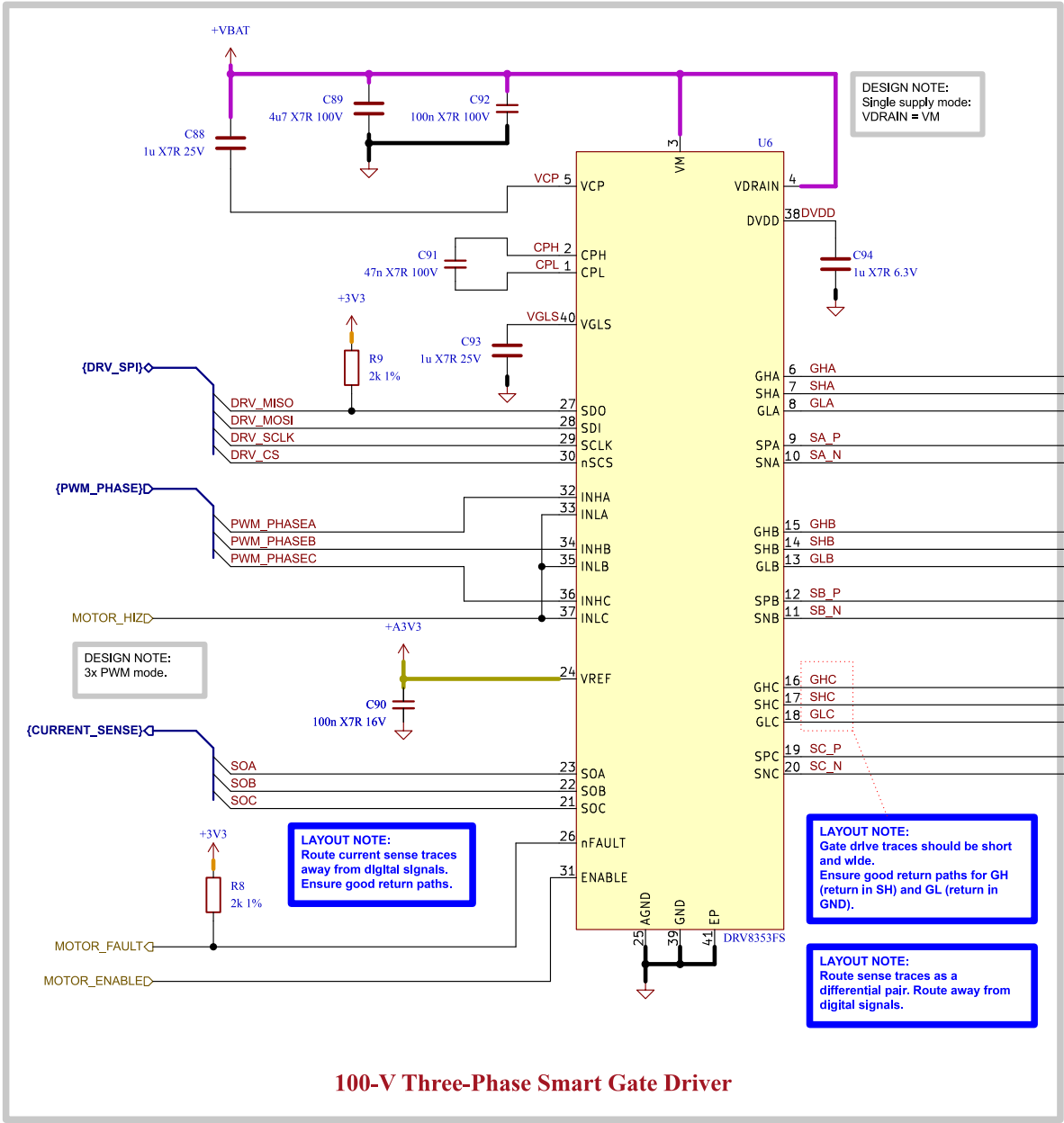
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V		Company: EPFL Xplore Research		Variant: Released	
Sheet Title: Power - Generation		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Path: /Project Architecture/Power - Generation/		File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
		Reviewer:		Size: A3	Sheet: 6 of 21

[7] Power - Connectors



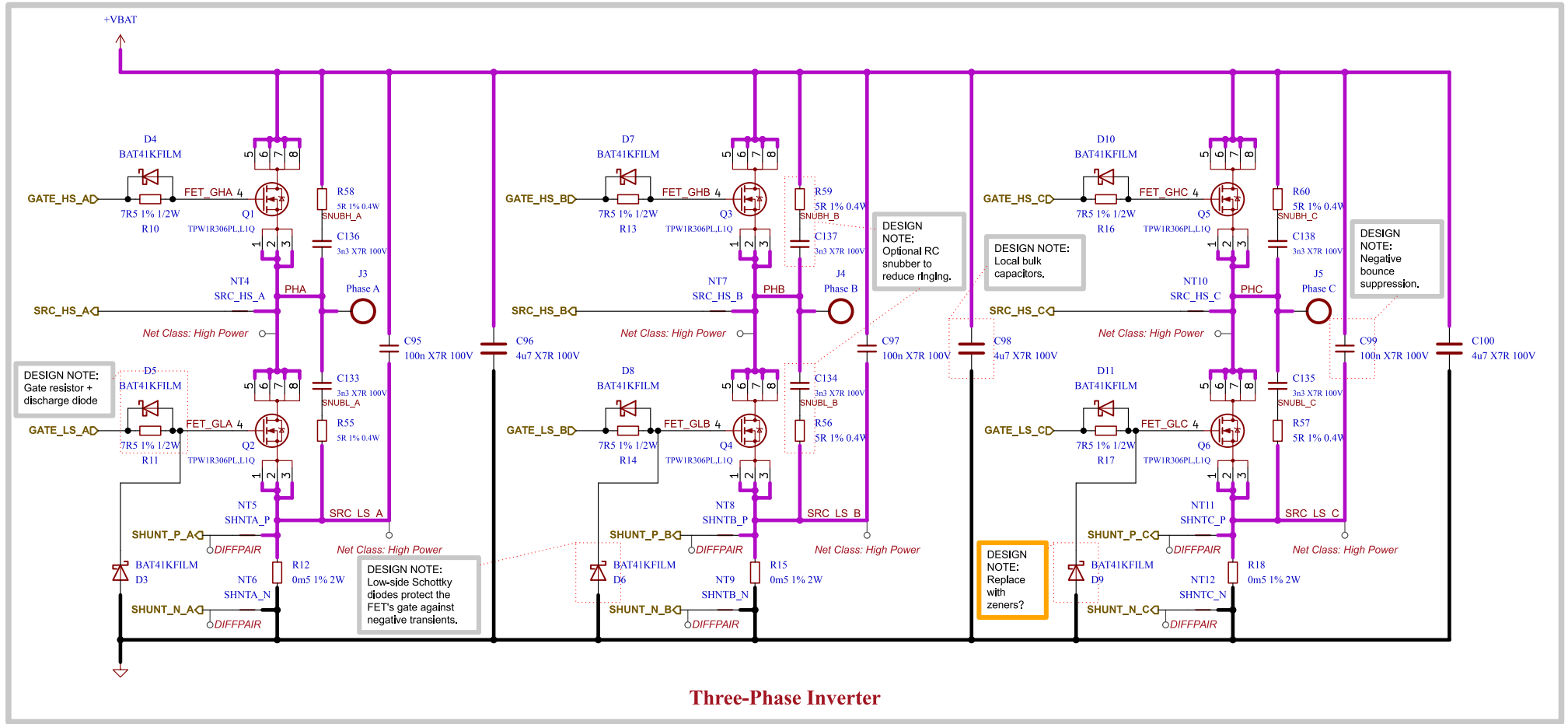
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		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Power - Connectors/	Reviewer:	Size: A4	Revision: 1.1
			Sheet: 7	of 21

[8] Motor Control - Top Level



	Comments:	Company: EPFL Xplore Research				Variant: Released	
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé			
	Sheet Title: Motor Control - Top Level	File Name: Motor Control - Top Level.kicad_sch	Designer: Vincent Nguyen		Date: 2023-12-20	Revision: 1.1	
	Sheet Path: /Project Architecture/Motor Control - Top Level/		Reviewer:		Size: A3	Sheet: 8 of 21	

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Reviewer:



Variant:
Released

Project Name:
Chienpanzé

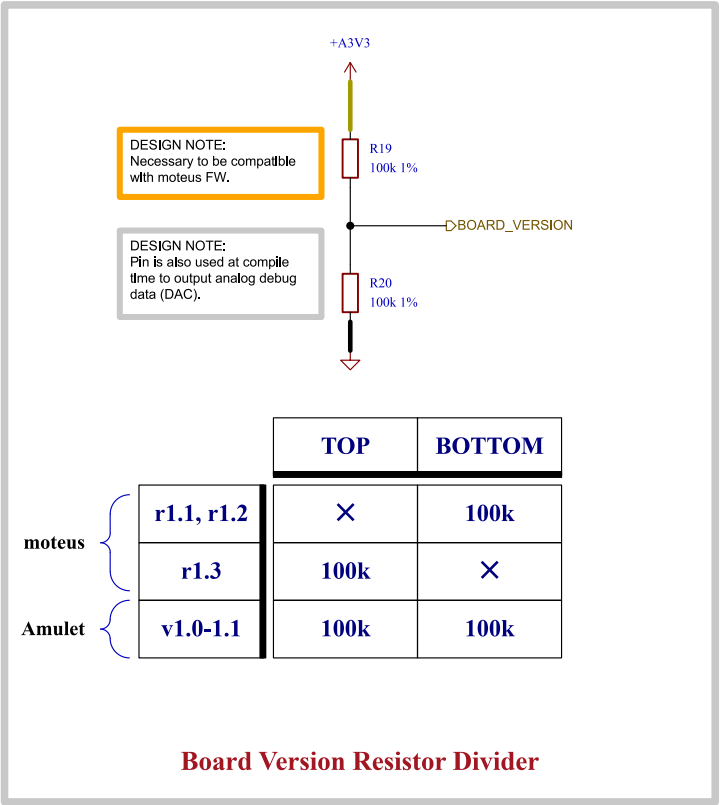
Date:
2024-01-25

Size:
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Revision:
1.1

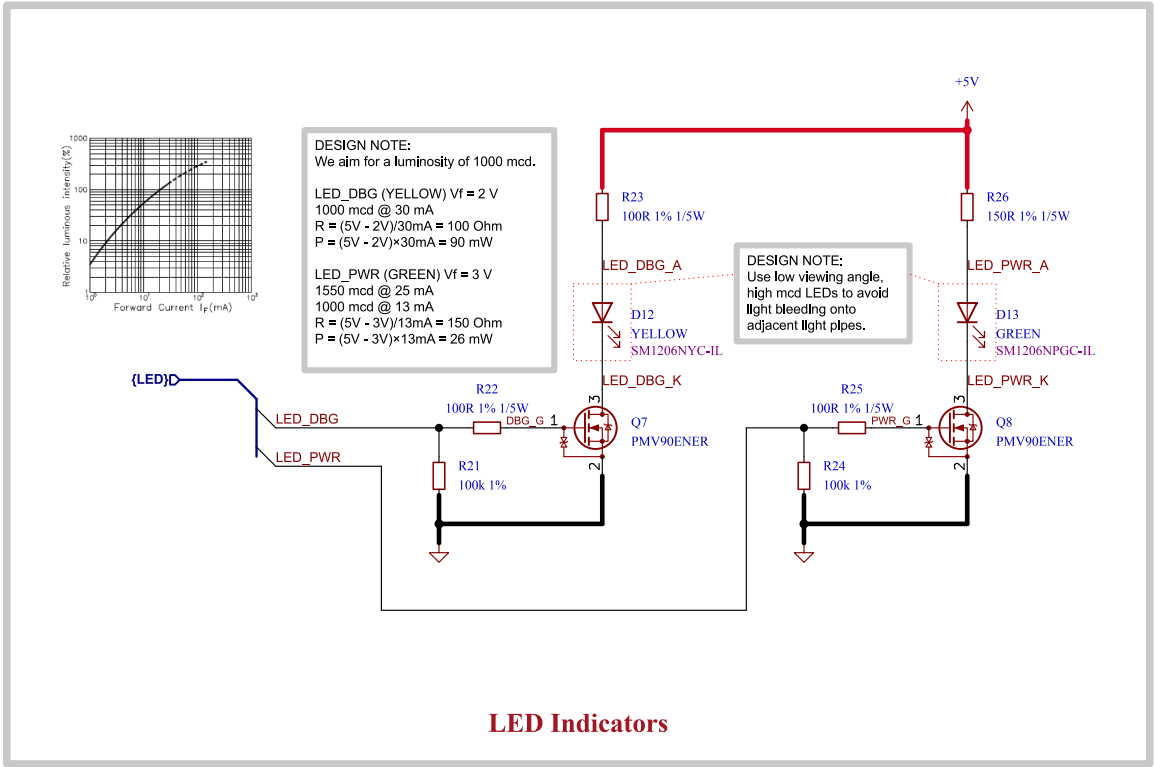
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9 of 21

[10] Misc - Board Version, DAC



	Comments:	Company: EPFL Xplore Research 		Variant: Released	
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 21

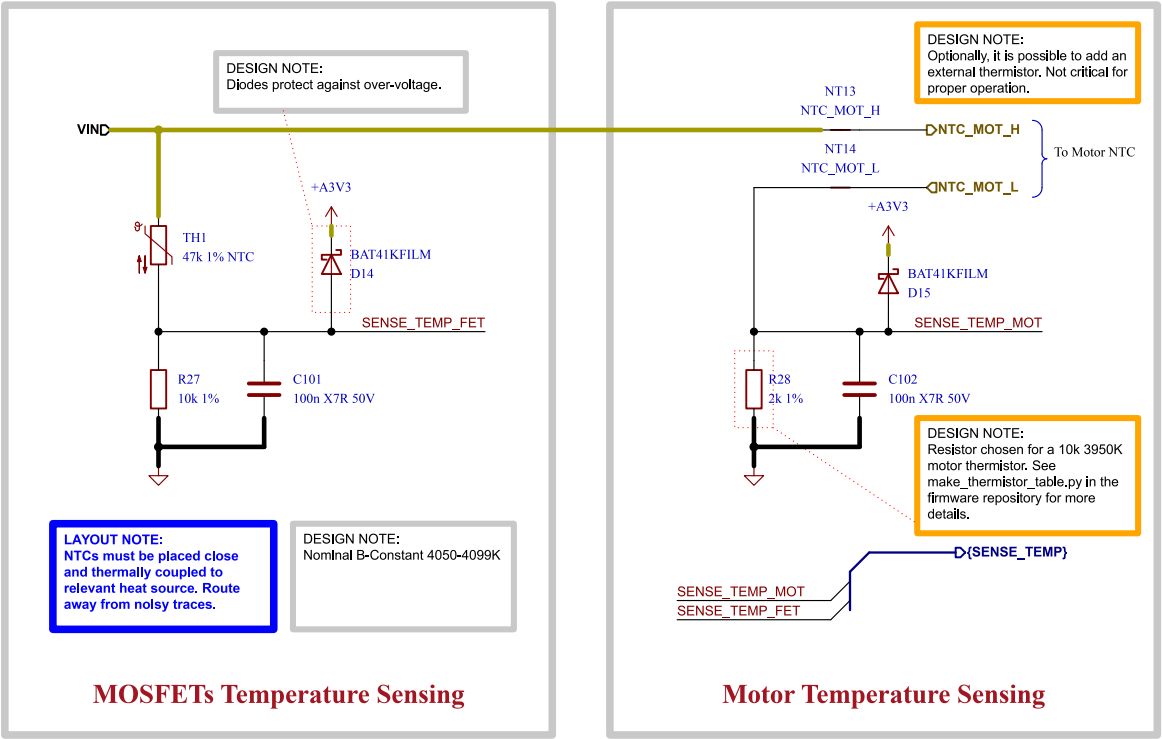
[11] User - LED Indicators



LED Indicators

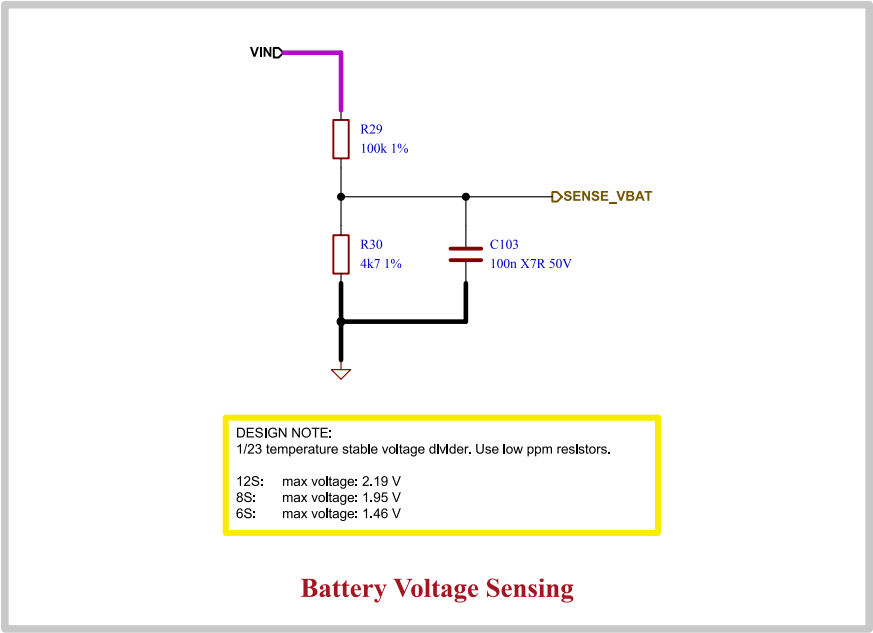
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	Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-19	Revision: 1.1
	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



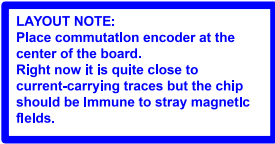
	Comments:	Company: EPFL Xplore Research		Variant: Released	
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer:	Size: A4	Sheet: 12 of 21

[13] Sensing - Battery

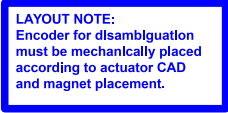


	Comments:		Company: EPFL Xplore Research		Variant: Released	
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	Sheet Title: Sensing - Battery		File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1
	Sheet Path: /Project Architecture/Sensing - Battery/			Reviewer:	Size: A4	Sheet: 13 of 21

A
B
C
D



DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



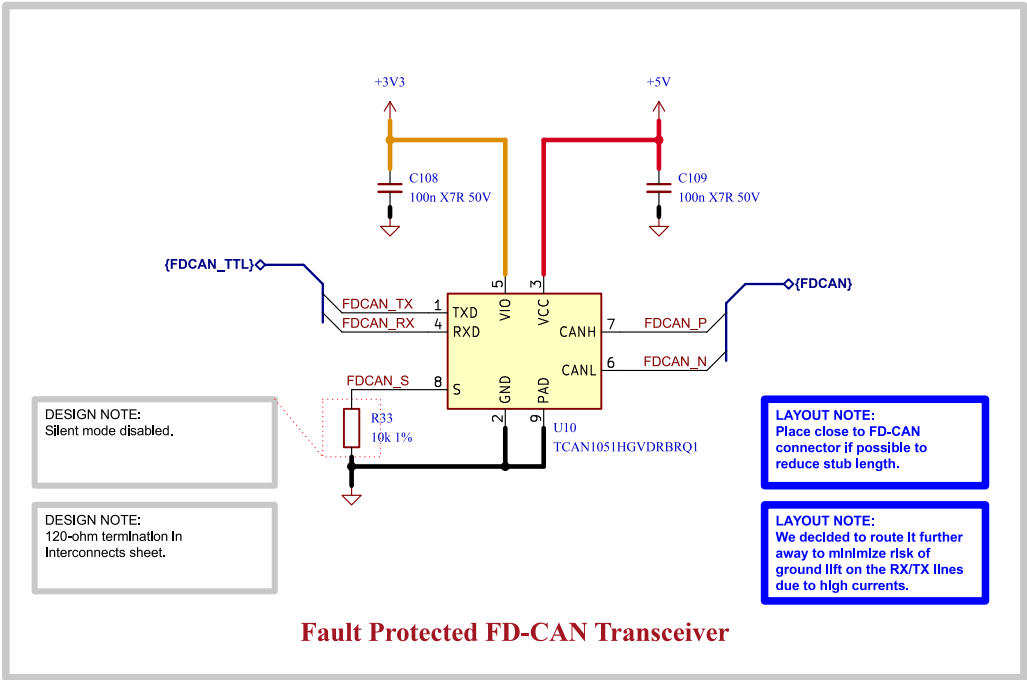
DESIGN NOTE:
The encoder for
disambiguation
cannot be used
to estimate
backlash!

D

[illegible]

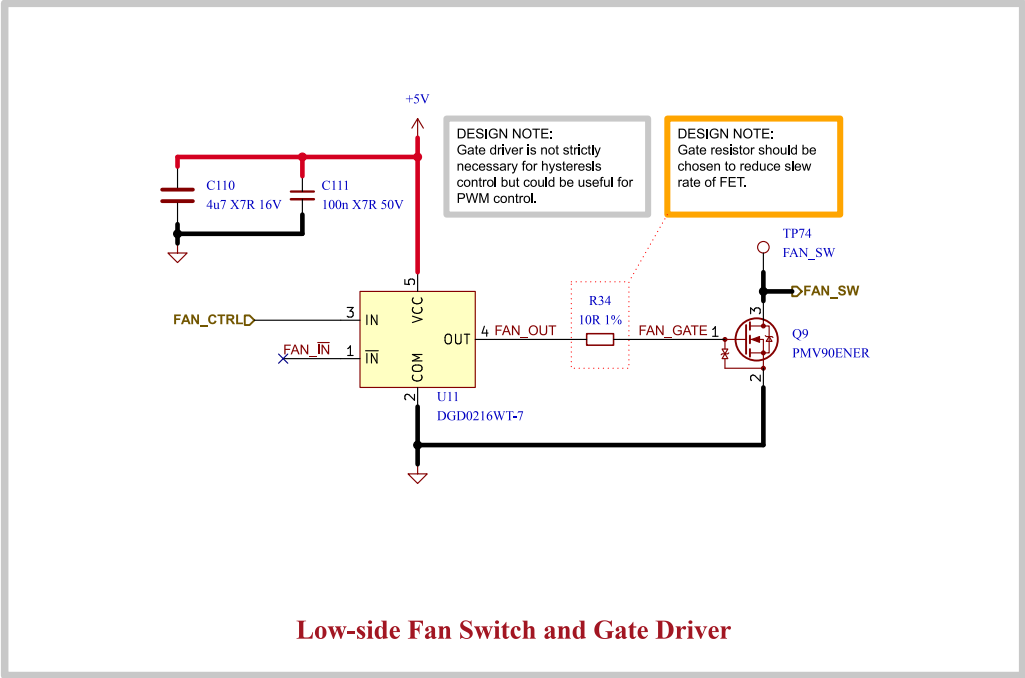
	Comments:	Company: EPFL Xplore Research				Variant: Released		
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé			
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen		Date: 2023-10-15		Revision: 1.1	
	Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:		Size: A4	Sheet: 15 of 21		

[16] Interface - FD-CAN



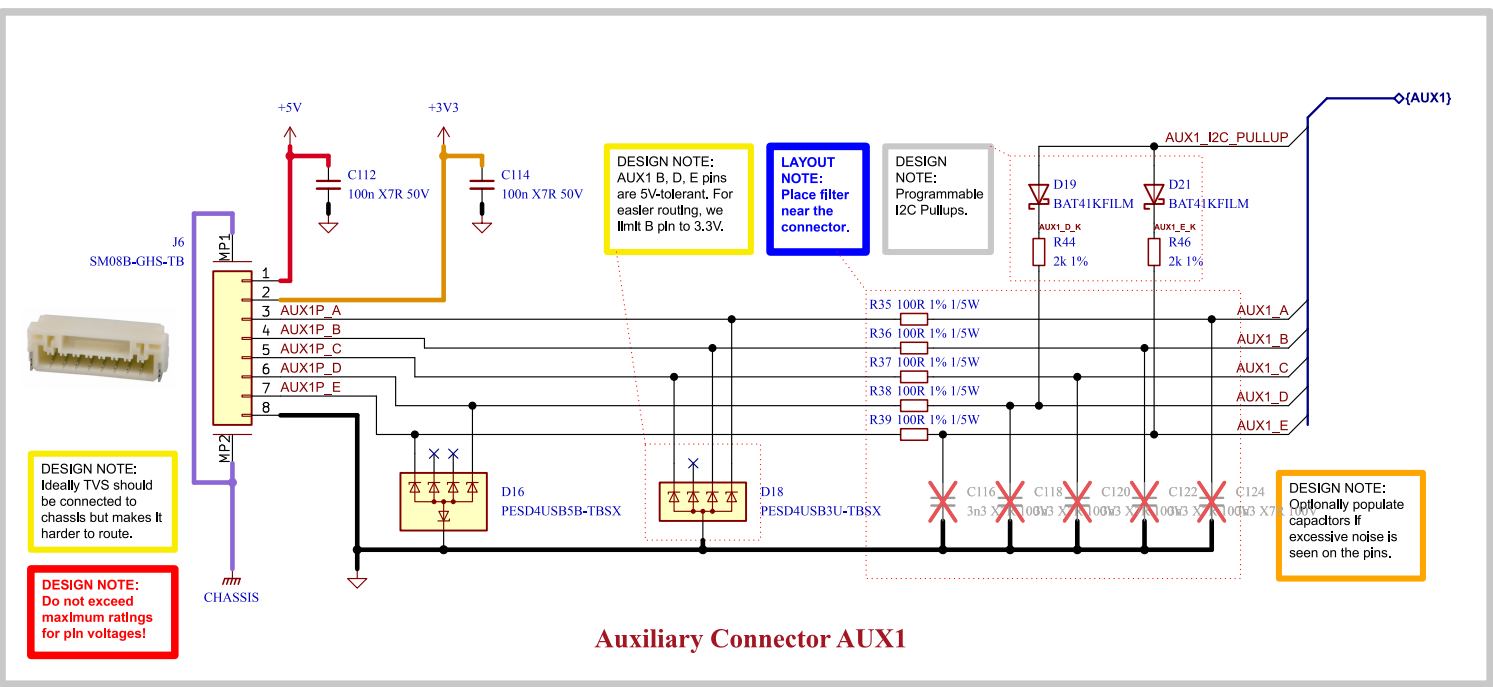
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			Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1	
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21	

[17] Interface - Fan Control

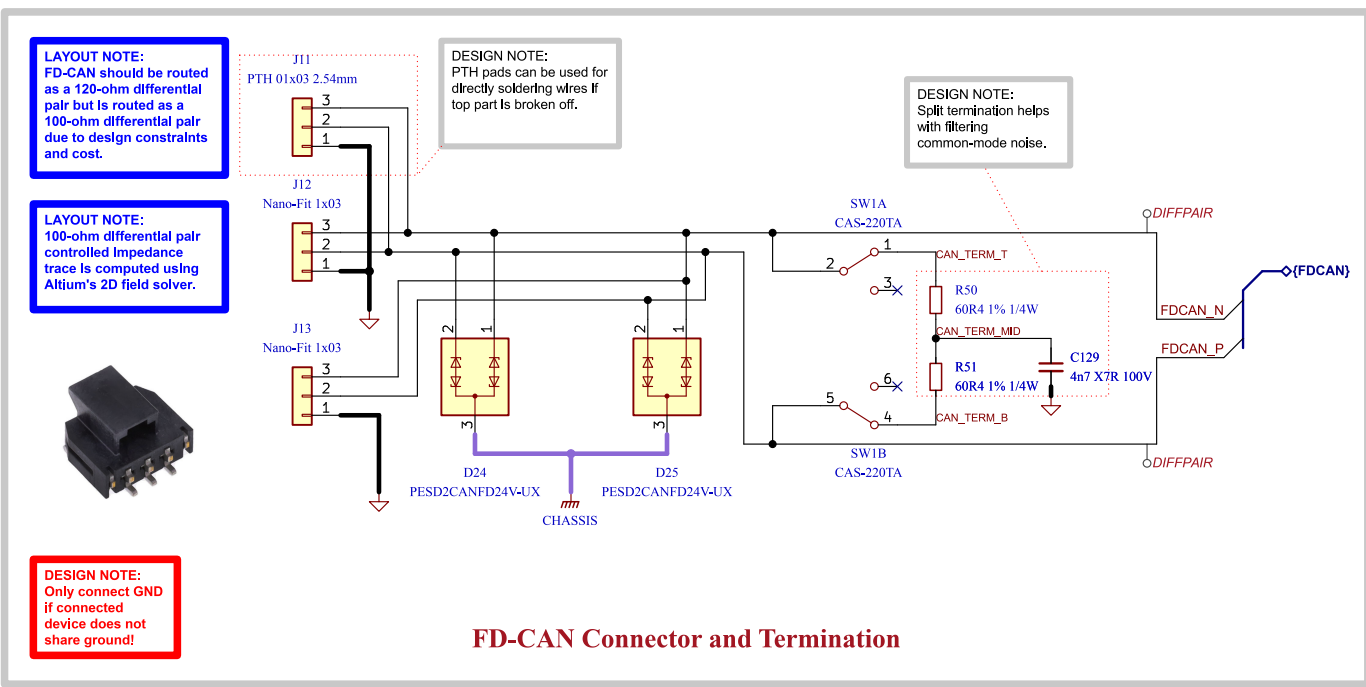


	Comments:	Company: EPFL Xplore Research		Variant: Released	
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen	Date: 2023-11-19	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21

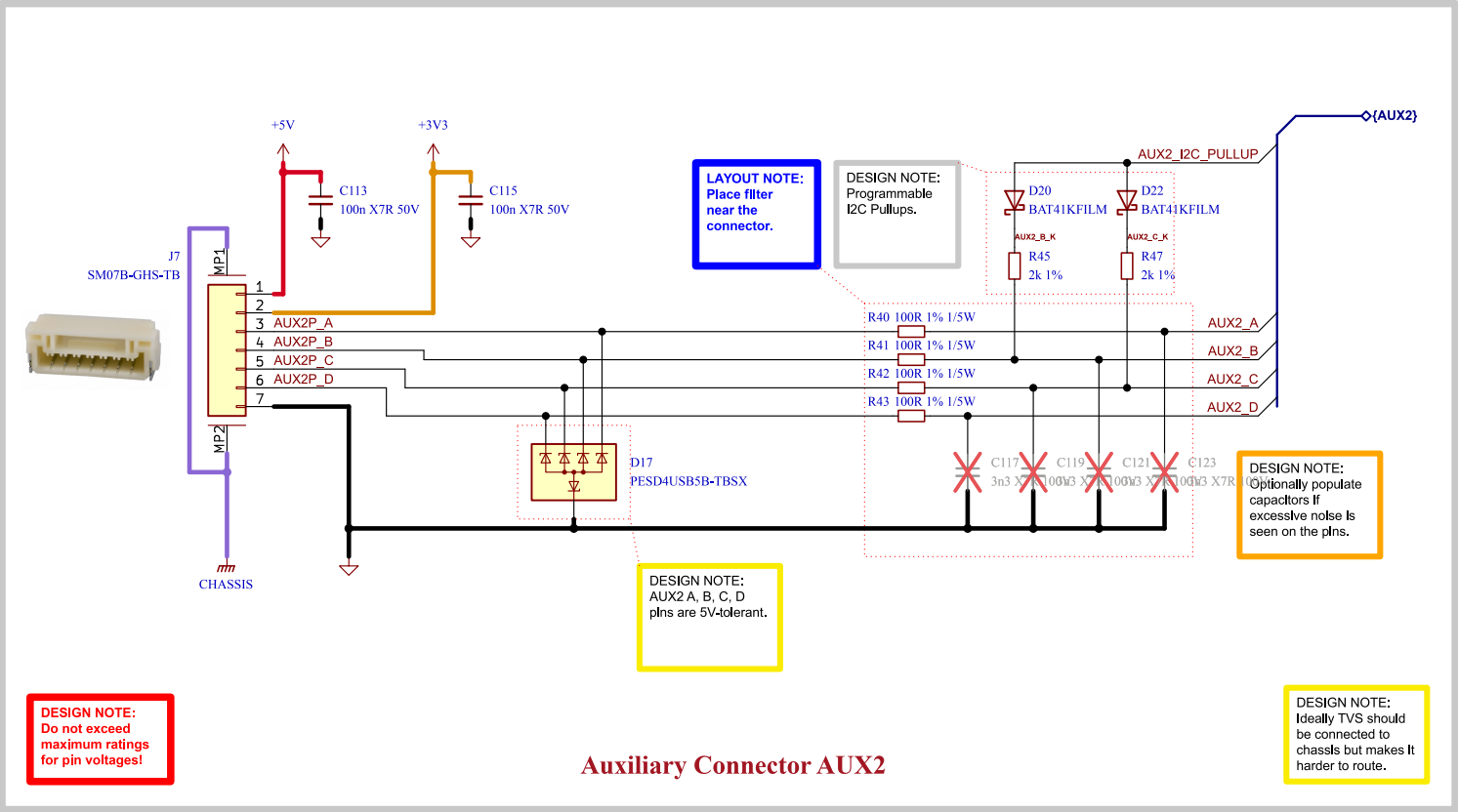
[18] Interface - Interconnects



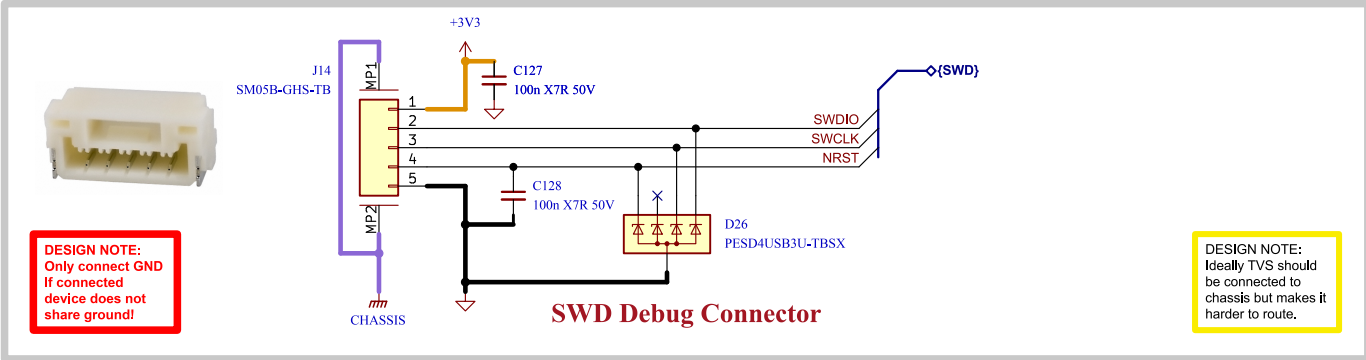
Auxiliary Connector AUX1



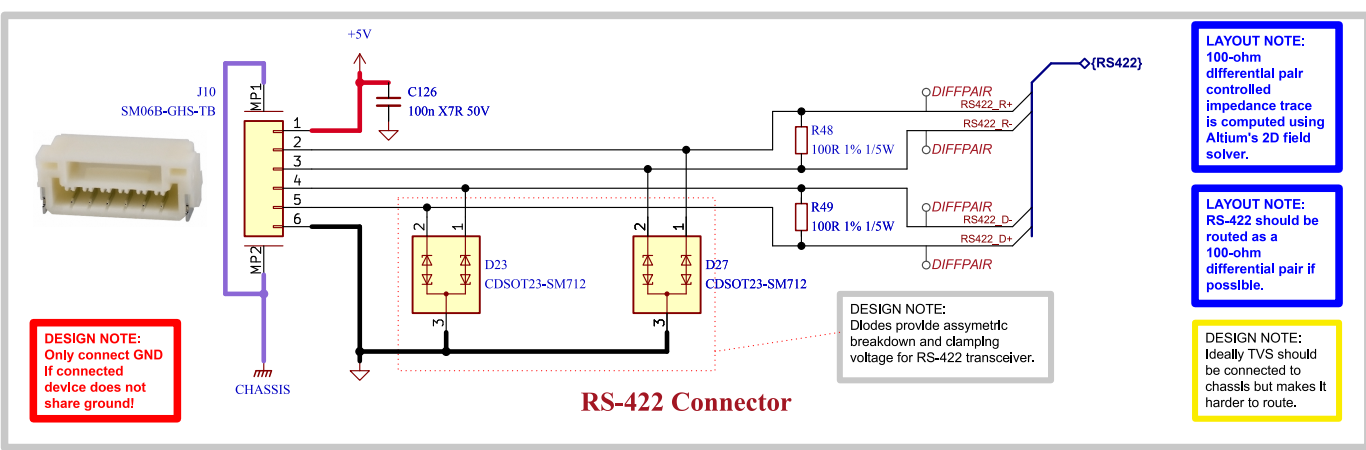
FD-CAN Connector and Termination



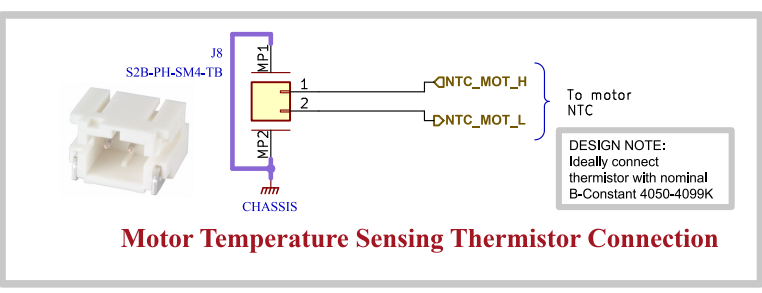
Auxiliary Connector AUX2



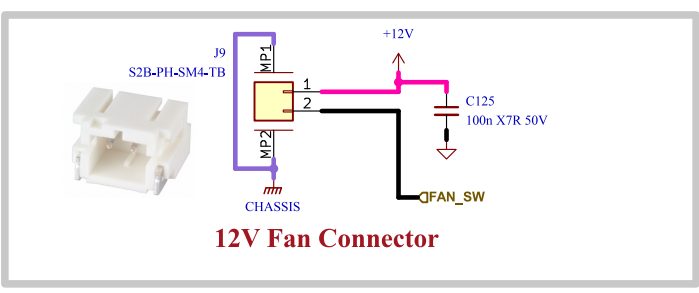
SWD Debug Connector



RS-422 Connector



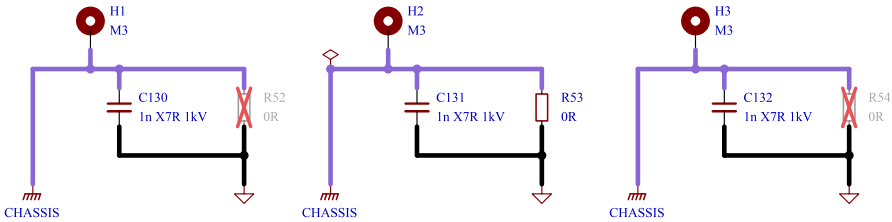
Motor Temperature Sensing Thermistor Connection



12V Fan Connector

Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant: Released	
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:		Revision: 1.1
			Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

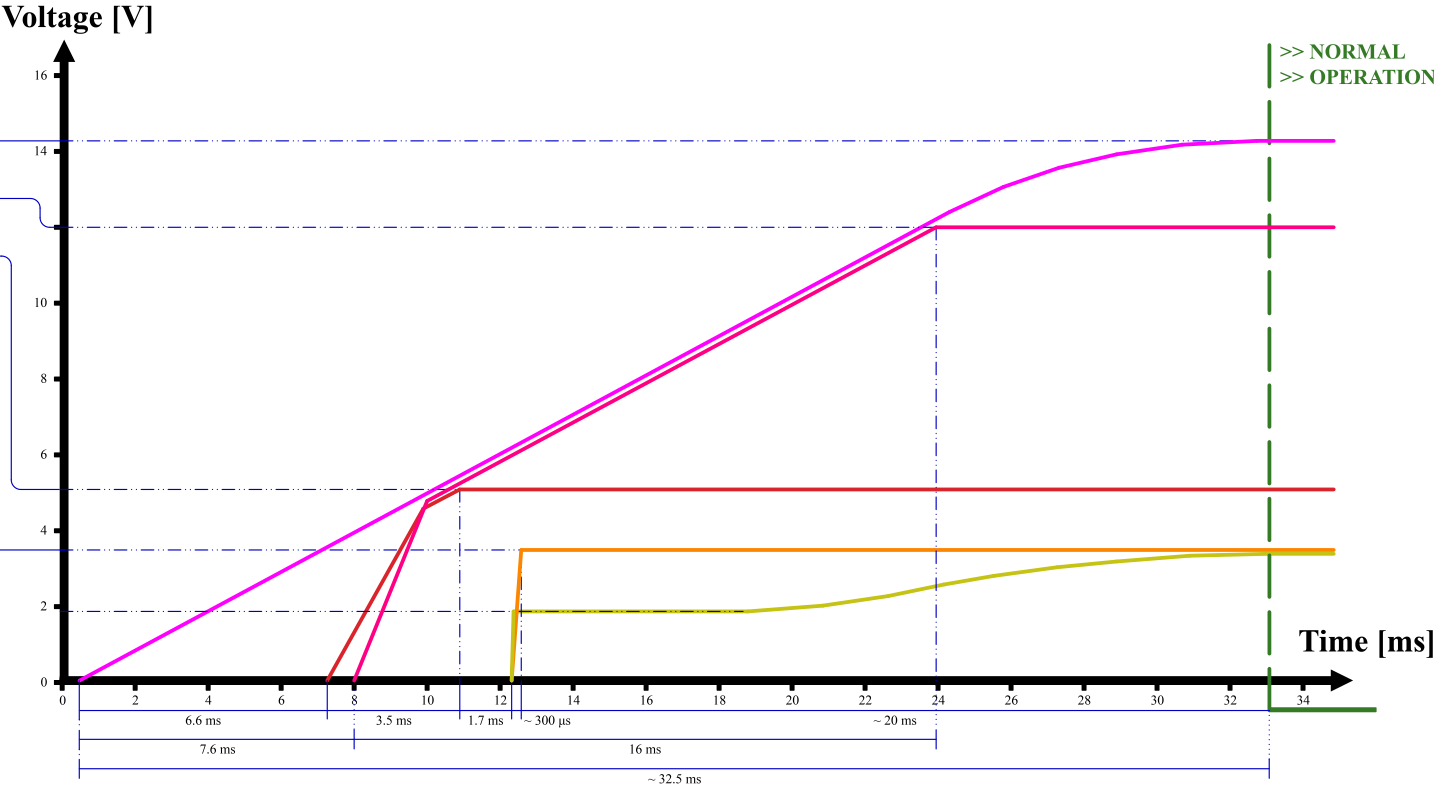
	Comments:		Company: EPFL Xplore Research		Variant: Released	
			Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1	
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21	

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research	Variant: Released	
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: 1.1
			Sheet: 20 of 21	

[illegible]